SORUPARAJAN T

Mobile Number: +91 8489924267
Location: Salem, Tamil Nadu
Mail id: soruparajanofficial@gmail.com
LinkedIn profile: www.linkedin.com/in/soruparajan-t
GitHub Account: https://github.com/soruparajan

PROFESSIONAL SUMMARY

Electronics and Communication Engineering student specializing in VLSI design with knowledge in analog circuits and digital circuit design. Strong foundation in CMOS devices & Finfet operations, transistor-level analog design, and digital logic systems including combinational and sequential circuits. Understanding of IC design principles such as device matching, parasitic effects, and design rule constraints. Focused on developing expertise in analog and mixed-signal integrated circuit development and physical design.

TECHNICAL SKILLS

- ◊ **LANGUAGES:** Verilog, C, Python,Perl.
- ◊ **EDA TOOLS:** Cadence Xcelium, Cadence Genus, Cadence Innovus, Cadence Pegasus, Virtuoso, Open-ROAD(7nm) , AMD Vivado, Quartus, Orange data mining tool.
- ◊ **CORE CONCEPTS:** CMOS Device Physics, Analog Circuit Fundamentals, Differential Pair & Current Mirror Design, Device Matching & Symmetry Techniques, Common Centroid & Interdigitation, Parasitic & RC Delay Analysis, DRC/LVS Verification Flow, Layout Dependent Effects (LDE), Static Timing Analysis (Basic).
- ◊ **BASIC TOOLS:** Git, GitHub, VS Code, Word, Colab.

PROJECTS

VLSI-Enabled AI Automation for Fault Detection and Predictive Maintenance in Aerospace Systems-45nm (Team project – 3 members)

- ✓ Architected a VLSI-based intelligent monitoring system for real-time fault detection.
- ✓ Integrated AI-driven predictive algorithms with hardware control logic.
- ✓ Designed and simulated RTL modules for system-level reliability enhancement.
- ✓ Analyzed trade-offs between performance, area, and reliability.
- ✓ Targeted aerospace-grade robustness and high-availability design considerations.

Two-Stage CMOS Operational Amplifier Design – Individual work

- ✓ Designed a two-stage CMOS operational amplifier with differential input stage and current mirror biasing to achieve high gain and stable amplification.
- ✓ Implemented Miller compensation to improve phase margin and ensure stable closed-loop operation.
- ✓ Simulated key performance metrics including DC gain, gain-bandwidth product (GBW), slew rate, phase margin, and power consumption.
- ✓ Performed Monte Carlo analysis to evaluate the impact of device mismatch and process variations on amplifier performance and reliability.

Low-Power Real-Time Speech Recognition System Integration (Team project – 3 members)

- ✓ Designed a hardware-efficient speech recognition architecture optimized for low-power real-time processing.
- ✓ Integrated signal preprocessing and feature extraction modules with control logic for embedded implementation.
- ✓ Analyzed performance, latency, and power trade-offs to ensure energy-efficient operation for edge application.

PATENT

Integrated Sensor and AI System for Adaptive Crowd Control in Open Environments

- ✓ Indian Patent Application No: 202641022240
- ✓ Published in Indian Patent Journal on 06 March 2026.
- ✓ Contributed to AI-based sensor fusion architecture for real-time crowd monitoring and emergency response.
- ✓ Co-Inventor among the listed inventors.

RESEARCH PUBLICATIONS

1. **VLSI-Enabled AI Automation for Fault Detection and Predictive Maintenance in Aerospace Systems**

✓ Presented at IAES 2026 Conference, ISRO Mahendragiri.
✓ Research focuses on AI-assisted fault prediction and VLSI-based monitoring architectures for aerospace applications.
✓ Conference paper accepted and presented.

2. **Low-Power Real-Time Speech Recognition System using AI**

✓ Presented at TQBCET'26 conference, Christ University Pune.
✓ Research Focuses on Real time speech recognition Using AI algorithms shown result through FPGA implementation
✓ Conference paper accepted and presented.

INTERNSHIP EXPERIENCE

Digital Electronics & VLSI Intern

Codec Technologies Pvt. Ltd. | Nov 2025 – Feb 2026

◊ Worked on RTL design, simulation, and synthesis using industry-standard Cadence tools.
◊ Developed and verified digital modules with focus on timing and logic optimization.
◊ Gained exposure to backend concepts including placement, routing, and PPA analysis.
◊ Built strong foundation in CMOS design principles and VLSI implementation flow.

EDUCATIONS

◊ **B.E. Electronics and Communication Engineering**
Mahendra Engineering College, Namakkal | 2022 – 2026 | CGPA: 7.9(Upto 6th semester)
◊ **Higher Secondary (Class XII) |** SRV Boys Higher Secondary School, Rasipuram ,2022 | Percentage: 78%

ACHIEVEMENTS

◊ Secured **Top 3%** (30/1250) **Rank** in the *AI/ML-Powered FPGA Design & Simulation Hackathon* conducted during Industry Academia Conclave – 2025 at **NIT Jamshedpur**, in collaboration with VLSI For All.
◊ Invited as **Resource Person** at Mahendra Engineering College to deliver a session on **"High Performance System Design using EDA Tools"** (9 March 2026), demonstrating Cadence-based VLSI design workflows.
◊ Secured **1st Prize** at INNOVAX-2025 (Mini Project Expo) for presenting the project *"Lung Cancer Detection"* organized by the Department of ECE, Mahendra Engineering College.

CERTIFICATIONS

◊ Participated in **"**VLSI Frontiers: Exploring Cadence Tools**"**, organized by the ECE Association, PSG Institute of Technology and Applied Research, Coimbatore (2025)
◊ VLSI for Beginners – 7-Day Training Program, VLSI For All (Jan 2026)
◊ NPTEL – Hardware Modeling using Verilog (2025)
◊ Training Program on Semiconductor Circuit Design and Simulation – Vellore Institute of Technology (VIT), Vellore (Sponsored by British Council), Jan 2026
◊ Cadence RTL-to-GDSII Flow v6.0 (2026)
◊ Cadence Digital Ic Fundamentals (2026)
◊ NPTEL – Digital Design With Verilog using Verilog (2026) – ELITE